Global Airbnb Market Audit: Strategic Executive Report

Project Overview

This audit analyses Airbnb’s market presence across five major global cities: London, Bangkok, Rome, Barcelona, and Amsterdam. By processing over 292,000 raw data points and refining them into 140,000 active listings, this report provides a clear picture of how the short-term rental market actually functions today.

The Four Problems Being Addressed

To understand this market, the research was broken down into four simple questions that explain Airbnb's impact:

1. **Market Composition:** To know if Airbnb is still a "home-sharing" site for individuals or if it has become a business platform. We distinguish between "Casual" hosts (renting out one home) and "Commercial" hosts (businesses running multiple listings).

2. **Geospatial Concentration:** To identify exactly which neighbourhoods have the highest density of rentals. This helps identify which areas are at risk of "over-tourism" and where traditional hotels are facing the most intense local competition.

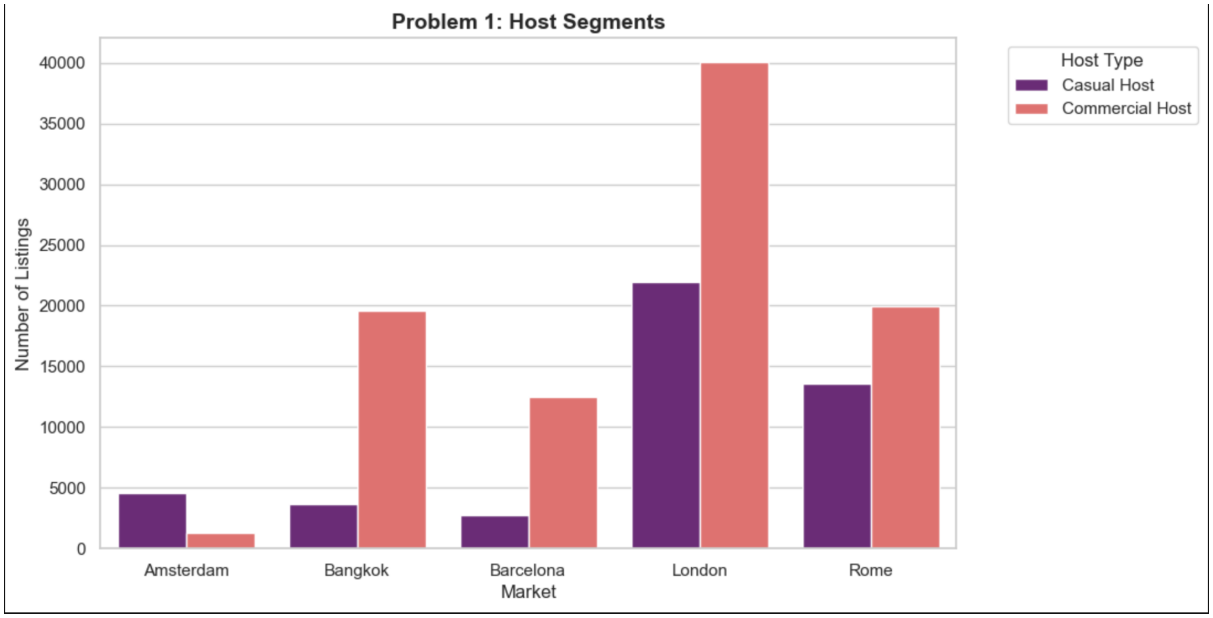
3. **Demand Velocity:** Just because a house is listed doesn't mean people are staying there. We look at how many reviews a listing got in the last year to see which cities and types of rooms are actually being booked right now.

4. **Regulatory Compliance:** Many cities now require hosts to have a license. We measure how many listings are "Verified" (have a visible license) versus "Unverified" (operating in a legal grey area) to see where the market is most at risk of being shut down by the government.

Key Insights and Analysis

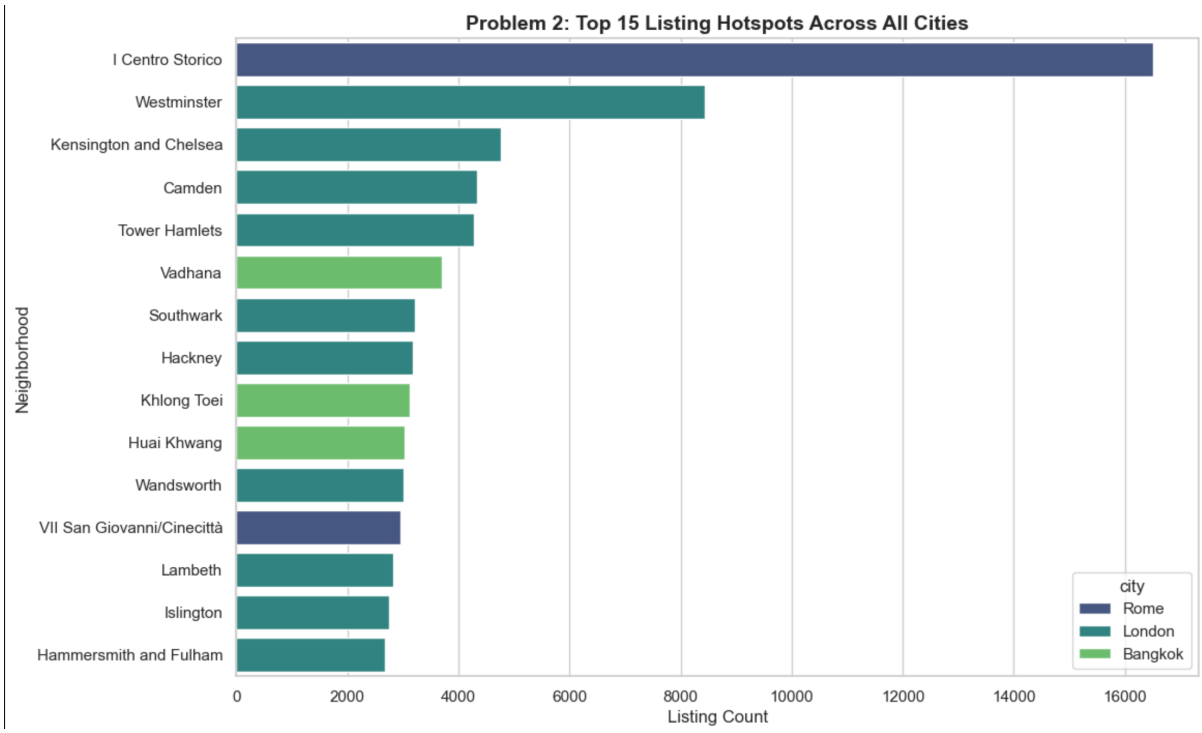
1. Market Composition

The data proves that Airbnb is no longer just for individuals. In cities like **London**, "Commercial" hosts manage over 40,000 listings. In **Amsterdam**, these professional hosts charge significantly more (averaging $549) than casual hosts ($275), showing that they are operating high-end, professionally managed hotel-style portfolios.



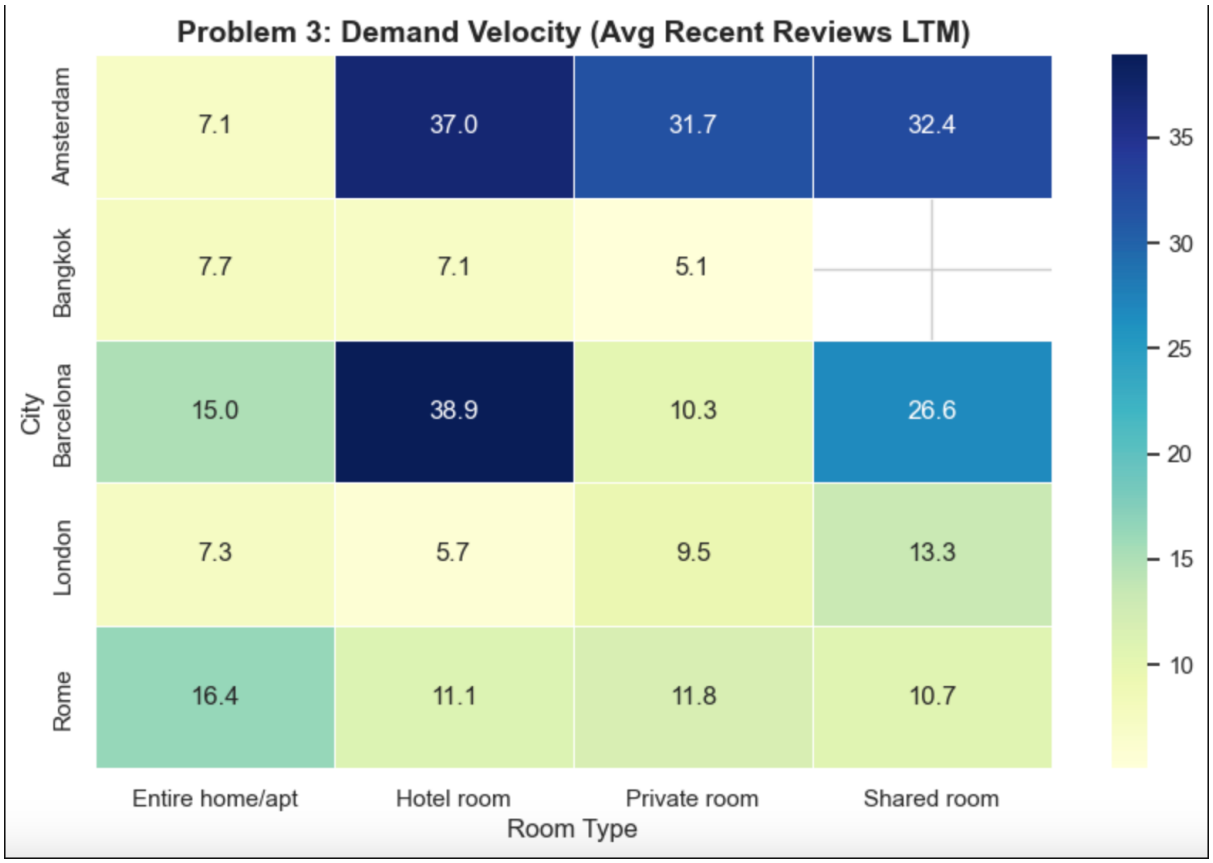
2. Geospatial Concentration

Certain neighbourhoods are incredibly crowded with rentals. **Rome's Historical Center (I Centro Storico)** is the most extreme example, with over 16,500 listings in one area. This creates a "hub" effect where Airbnb supply significantly outweighs traditional hotel rooms, putting pressure on local infrastructure.
When thousands of short-term rentals are packed into one area, the neighbourhood stops being a place for residents and becomes a "tourist factory." The city's infrastructure - like trash collection, water, and power - breaks down because it was designed for stable families, not a constant rotation of thousands of visitors. Ultimately, this forces out local shops and neighbours, turning a vibrant community into a hollowed-out hub that serves only temporary guests.



3. Demand Velocity

By looking at recent reviews, we found that "Hotel Rooms" listed on Airbnb in **Barcelona** and **Amsterdam** are actually the most active listings. This means that traditional boutique hotels are now using Airbnb as their primary way to find customers, rather than just independent homeowners.



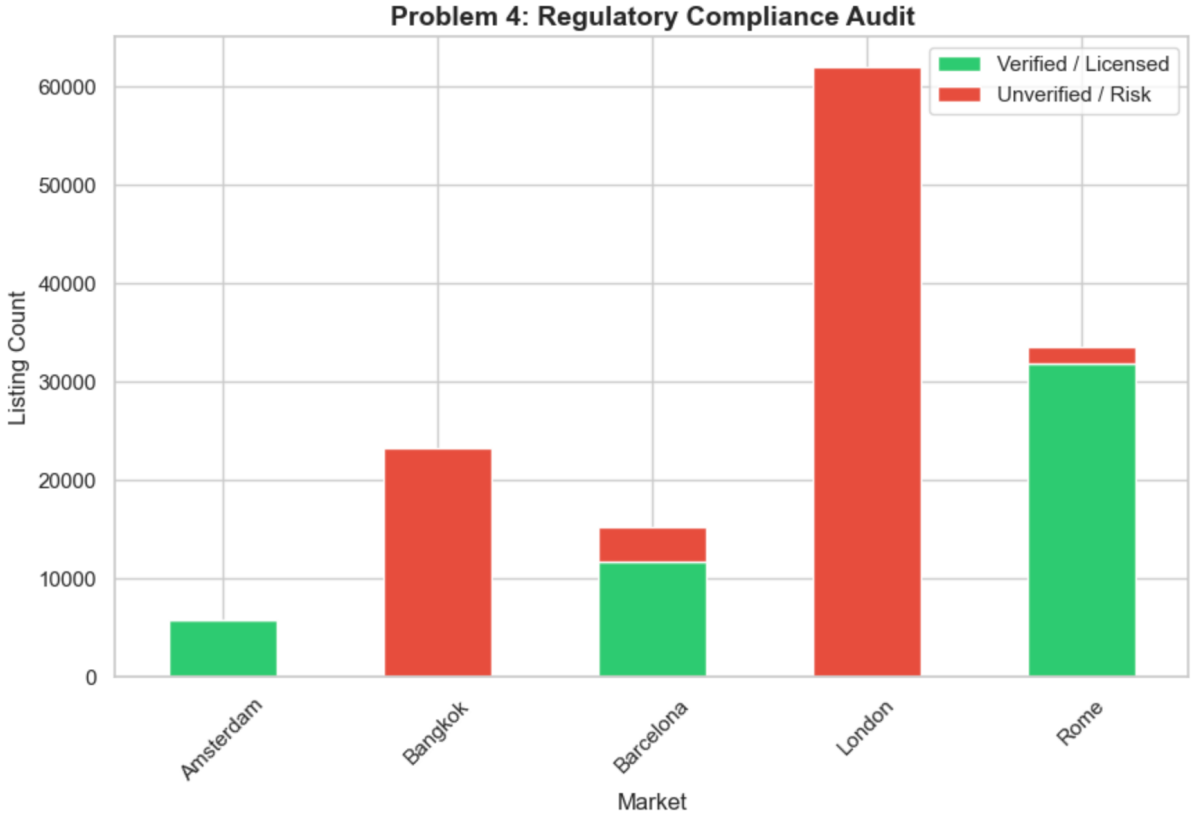
Shared rooms represent an insignificant or non-active market segment in Bangkok based on the filtered dataset.

4. Regulatory Compliance

There is a massive "Risk Gap" between cities. **Amsterdam and Barcelona** are highly regulated and transparent. However, **London and Bangkok** show almost zero visible licensing in the data. This makes London a "Grey Market"—if the city suddenly decides to enforce strict license displays, over 60% of the listings could disappear instantly, causing a massive supply shock.

This is further confirmed via secondary research that Airbnb listings in London and Bangkok are not uniformly verified via government license, though both cities have strict, differing regulations. Many rentals operate illegally without licenses, requiring

guests to verify compliance to avoid booking cancellation or disruption by authorities.



Strategic Recommendations

- **For Investors:** Focus on **Amsterdam** for long-term stability due to its mature legal environment. Be cautious with **London** and **Bangkok**, as it is a high-growth but high-risk market.

- **For Hotel Managers:** In R**ome**, your biggest competition is concentrated in the historical center. Tailor your pricing specifically to compete with the 16,000+ alternative listings in that district.

- **For Future Research:** Don't just count the total number of houses; look at how many reviews they get. A "busy" market is a much better indicator of value than a "large" market with many empty rooms.

Data Quality Statement

This analysis was performed after a careful and thorough data cleaning/refinement of the this dataset. Listings with $0 prices, fake addresses, or those meant for long-term residential leases (over 365 days) were removed. This ensures the results reflect the **real competitive market** for travellers.
The initial dataset consisted of 292,000 raw records, which were reduced to 140,000 high-integrity listings through a rigorous SQL pruning process. A price filter removed entries priced below 10 dollars to eliminate test data and system errors; these values often occur when hosts input nominal amounts during initial account setup to verify

that a listing page is live. Additionally, a minimum nights filter excluded properties requiring stays of over 365 days to ensure the analysis focused exclusively on short-term travel inventory. Such high minimums are often used by hosts as branding placeholders, result from configuration mistakes, or signify an intent to advertise long-term residential leases rather than transient visitor accommodations. Finally, SQL CASE statements flagged missing license information as unverified to facilitate clear regulatory risk assessment. This systematic cleansing process ensured that the final dataset accurately represents the active and competitive market environment for travelers.